ZZ–C–101a
4 JANUARY 1951
SUPERSEDING
Fed. Spec. ZZ–C–101
November 7, 1933

FEDERAL SPECIFICATION

CATHETERS, URETHRAL, RUBBER

This specification was approved on the above date by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. CLASSIFICATION

1.1 Types and styles.—Urethral catheters covered by this specification shall be of the following types and styles, as specified:

Type I.—Plain.
  Style 1.—Robinson.
  Style 2.—Nelaton.
Type II.—Balloon.
  Style 3.—Hemostatic.
  Style 4.—Self-retaining.
Type III.—Malecot.
Type IV.—Pezzer.

1.2 Compositions.

1.2.1 *Type I and type III* catheters shall be of the following compositions, as specified:

Composition A.—Natural latex compound.
Composition B.—Natural rubber compound.

1.2.2 *Type II and type IV* catheters shall be of composition A, natural latex compound only.

2. APPLICABLE SPECIFICATIONS

2.1 The following Federal Specification, of the issue in effect on date of invitation for bids, forms a part of this specification:

ZZ–R–601—Rubber Goods; General Specifications (Methods of Physical Tests and Chemical Analyses).

(Copies of Federal specifications and the Federal Specifications Index may be obtained upon application, accompanied by postal note, money order, coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Prices may be obtained from the Superintendent of Documents.)

2.2 Specifications applicable only to individual departments are listed in section 7.

3. REQUIREMENTS

3.1 Material.

3.1.1 *Composition A.*—Composition A shall be a natural latex compound.

3.1.2 *Composition B.*—Composition B shall be a natural rubber compound.

3.2 Description.—The catheters shall be homogenous throughout.

3.3 Sizes, dimensions, and physical properties.—The sizes, dimensions, and physical properties of the catheters for the various types shall be as shown in tables I, II, III, and IV. A tolerance of plus or minus 1 Fr. (0.3 mm.) in outside diameter is acceptable. The diameter expressed in millimeters according to the French scale is numerically equal to one-third the scale size.

3.4 Ultimate elongation.

3.4.1 *Composition A.*—The ultimate elongation before aging shall be not less than 800 percent (4.3.1).

3.4.2 *Composition B.*—The ultimate elongation before aging shall be not less than 400 percent (4.3.1).

3.5 Resistance to aging.—The rigidity of the catheters shall not change more than 25 percent, and the breaking strength and elongation shall not decrease more than 25 percent, nor shall the rubber crack or become tacky or brittle, when subjected to the accelerated aging test described in 4.3.3.

3.6 Color.—Unless otherwise specified, the catheters shall be amber, natural, or red.

3.7 Identification marking.—Each catheter shall be plainly and permanently marked upon the proximal end with the name or registered trade-mark of the manufacturer, the date of

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manufacture (month and year), and the French size.

TABLE I.—*Sizes and physical properties for type I catheters*

French size	Inside diameter,[1] minimum	Rigidity [2]				Breaking strength, minimum
		Style 1, Robinson		Style 2, Nelaton		
		Minimum	Maximum	Minimum	Maximum	
	Inch	*Grams*	*Grams*	*Grams*	*Grams*	*Pounds*
8	0.040	3	7	6	10	12
10	.040	6	12	8	14	12
12	.057	8	14	8	14	16
14	.070	14	26	16	28	20
16	.098	20	38	24	42	28
18	.112	39	76	39	76	36
20	.140	60	120	60	120	44
22	.154	85	170	85	170	52
24	.182	100	200	100	200	60
26	.190	110	250	110	250	68

[1] Measured at any point.

[2] Rigidity requirements applicable to composition B only.

3.8 Type I, plain.—Type I catheters shall be plain and shall be similar in design to those shown in figure 1. The catheters shall be not less than 15 inches in length.

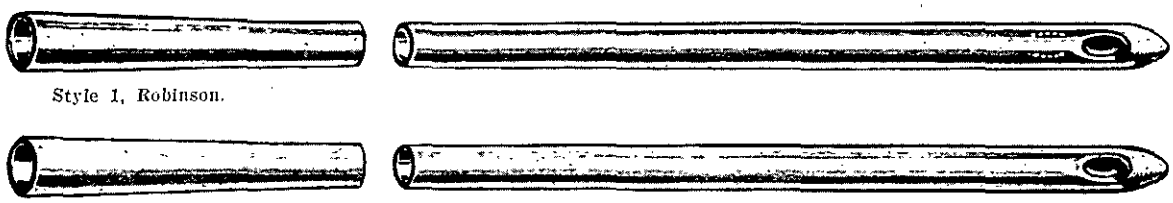
3.8.1 *Style 1, Robinson.*—Style 1 catheters shall be of the Robinson design with two eyes and a hollow tip. The proximal end shall be funnel-shaped. The tip on sizes 16 through 26, of catheters fabricated of composition A, shall be reinforced. The characteristics of composition B shall be such that reinforcement will not be required. The eyes shall not overlap, each eye shall be oval-shaped, and the area of each eye shall be at least equal to the cross-sectional area of the lumen of the catheter proper.

3.8.2 *Style 2, Nelaton.*—Style 2 catheters shall be of the Nelaton design with a solid rounded tip and a depressed eye at the outlet. The proximal end shall be funnel-shaped. The eye shall be oval-shaped and the diameter of the bore connecting the eye to the lumen shall measure at least three-fourths of the diameter of the lumen of the catheter proper.

3.9 Type II, balloon.—Type II catheters shall be of the balloon type, and shall be similar in design to those shown in figure 2. The diameter of the balloon section of the catheters shall not exceed the diameter of the catheter (see 3.3) by more than 1 millimeter. The catheters shall be not less than 15 inches in length.

3.9.1 *Style 3, hemostatic.*—Style 3 catheters shall be hemostatic urethral catheters of the balloon type, self-retaining, with a bag of 30-cubic-centimeter capacity and a hollow reinforced tip. Two eyes shall be located in the end as shown in figure 2. The bag shall be made an integral part with the outer wall of the catheter with an inflation channel which shall be imbedded or permanently attached within the wall. Each eye shall be oval-shaped and the area of each eye shall be at least equal to the cross-sectional area of the lumen of the catheter proper.

3.9.2 *Style 4, self-retaining.*—Style 4 catheters shall be of self-retaining balloon design with a bag of 5-cubic-centimeter capacity, and a hollow reinforced tip on sizes 16 Fr. and larger. Two eyes shall be located in the sides of the end as shown in figure 2. The bag shall be made an integral part with the outer wall of the catheter with an inflation channel which shall be imbedded or permanently attached within the wall. The eyes shall be oval-shaped and the area of each eye shall be at least equal to the cross-sectional area of the lumen of the catheter proper.



Style 1, Robinson.

Style 2, Nelaton.

FIGURE 1.—*Type I, plain catheters.*

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TABLE II.—*Sizes and physical properties for type II catheters*

French size	Inside diameter,[1] minimum	Breaking strength, minimum
	Inch	*Pounds*
12	0.067	16
14	.084	20
16	.084	28
18	.098	36
20	.112	44
22	.140	52
24	.171	60
26	.197	68
28	.211	74
30	.225	82
32	.252	90

[1] Measured at any point.

3.10 Type III, Malecot.—Type III catheters shall be of the four-wing Malecot type, self-retaining with reinforced tips, and similar to the design shown in figure 3, and the outside diameter (over-all spread of diametrically opposite retaining wings) of the self-retaining device for the various sizes shall be as shown in table III. The proximal end shall be funnel-shaped. The catheters shall be not less than 13 inches in length.

TABLE III.—*Sizes and physical properties for type III catheters*

French size	Inside diameter,[1] minimum	Self-retaining device (outside diameter)		Rigidity [2]		Breaking strength, minimum
		Minimum	Maximum	Minimum	Maximum	
	Inch	*Millimeters*	*Millimeters*	*Grams*	*Grams*	*Pounds*
12	0.057	12	18	8	14	16
14	.070	12	18	16	28	20
16	.098	14	18	24	42	28
18	.112	14	19	39	76	36
20	.140	18	20	60	120	44
22	.154	18	22	85	170	52
24	.182	18	25	100	200	60
26	.190	21	29	110	250	68
28	.203	21	31	118	290	74
30	.210	21	33	124	340	82
32	.236	21	35	130	380	90
34	.239	21	38	136	420	95
36	.256	21	38	142	460	105
38	.268	21	40	148	500	115
40	.291	21	40	154	540	125

[1] Measured at any point.

[2] Rigidity requirements applicable to composition B only.

3.11 Type IV, Pezzer.—Type IV catheters shall be of the Pezzer type, with self-retaining raised top similar in design to that shown in figure 4. The proximal end shall flare slightly. The distal end shall have as an integral part of the catheter a self-retaining device which consists of an expansion or mushroom top of thin, easily collapsed walls, 3 ± 1 openings distally and a raised top. The catheters shall be not less than 13 inches in length.

TABLE IV.—*Sizes and physical properties for type IV catheters*

French size	Inside diameter,[1] minimum	Self-retaining device (outside diameter)		Breaking strength, minimum
		Minimum	Maximum	
	Inch	*Millimeters*	*Millimeters*	*Pounds*
12	0.057	8	17	16
14	.070	12.5	17	20
16	.098	13	17	28
18	.112	14	17	36
20	.140	15	20	44
22	.154	17	21	52
24	.182	19	22	60
26	.190	19	26	68
28	.203	19	26	74
30	.210	19	26	82
32	.236	19	32	90
34	.239	30	34	95
36	.256	30	34	105
38	.268	30	34	115
40	.291	30	34	125

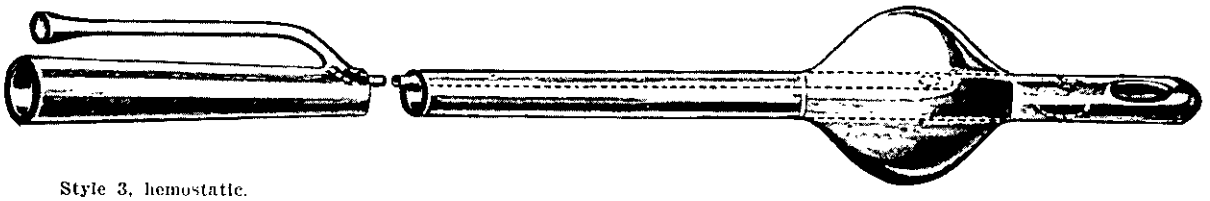
[1] Measured at any point.

3.12 Compatibility.—Catheters shall be fabricated of such material as to have no injurious or irritating effects on the body when used for the purpose intended.

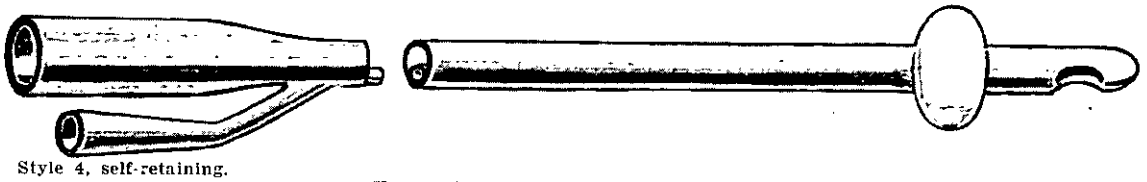
3.13 Workmanship.—The workmanship

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shall be first class throughout. The finished catheters shall be symmetrical in shape and the surface shall have a smooth finish, free from irregularities or imperfections of any kind which detract from their appearance or which may impair their serviceability.



Style 3, hemostatic.



Style 4, self-retaining.

FIGURE 2.—*Type II, balloon catheters.*

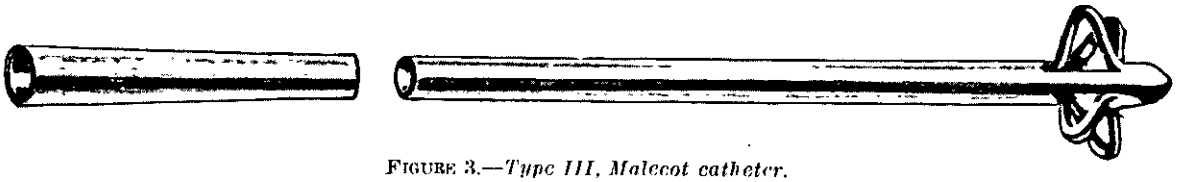


FIGURE 3.—*Type III, Malecot catheter.*

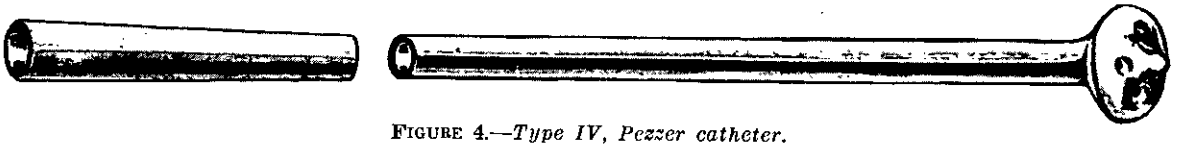


FIGURE 4.—*Type IV, Pezzer catheter.*

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.—Unless otherwise specified, not less than 6 catheters shall be taken at random from each 200 or less of each type, style, composition, and size delivered.

4.2 Inspection.—The samples shall be inspected for material, workmanship, markings, dimensions, finish, type, style, construction, color, and packaging.

4.3 Tests.

4.3.1 *Composition, ultimate elongation, breaking strength, and inside diameter.*—Unless otherwise specified, the tests for composition, ultimate elongation, breaking strength, and inside diameter shall be conducted in accordance with Federal Specification ZZ–R–601.

4.3.2 *Rigidity.*—A perforated rubber stopper shall be clamped in a suitable device so that the lower surface of the stopper is exactly 4 inches above the pan of a balance, when the pan is clear down. The catheter shall be passed through the stopper until the tip touches the pan when the pan is clear down. The rigidity is noted as the weight in grams which is required to bring the pan to a position just above an exact level. In performing the test, any given weight shall be applied for a period of 5 seconds and the weight removed between tests. Three catheters of each size delivered shall be tested and the rigidity for each size shall be considered to be the average of the three readings obtained. The rigidity both before and after aging shall be determined on the same catheter.

4.3.3 *Accelerated aging.*—The catheters shall be subjected to either of the following accelerated aging tests.

4.3.3.1 *Oxygen bomb.*—The catheters shall

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be subjected to the oxygen bomb test described in Federal Specification ZZ–R–601 for a period of 4 days at 70° ± 1° C. with oxygen pressure at 300 ± 10 pounds per square inch.

4.3.3.2 *Air oven.*—The catheters shall be subjected to the air oven test described in Federal Specification ZZ–R–601 for a period of 7 days at 70° ± 1° C.

5. PREPARATION FOR DELIVERY

5.1 Packaging.—Unless otherwise specified, each catheter shall be packaged in accordance with accepted commercial practice.

5.2 Packing.—The subject commodity shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation.

5.3 Marking.

5.3.1 *Packages.*—Unless otherwise specified, each package shall be marked with the name of the commodity, type, style, composition, size, and quantity contained therein, and the names of the contractor and manufacturer.

5.3.2 *Shipping containers.*—Unless otherwise specified, shipping containers shall be marked with the name of the commodity, the type, style, composition, and size as defined by the contract or order under which the shipment is made, the quantity contained therein, the name of the contractor, the number of the contract or order, the name and address of the consignee and the gross weight.

6. NOTES

6.1 Ordering data.—Purchasers should specify type, style, composition, and size required and should exercise any desired options offered herein. (See 1.1, 1.2, 3.6, 4.1, 5.1, 5.3, and 7.2.5.)

6.2 This specification covers the following items listed in the Armed Services Catalog of Medical Matériel: 3–174–200 through 3–174–300 for Catheters, Urethral, Rubber, Robinson: 3–175–500 through 3–175–510 for Catheters, Urethral, Rubber Hemostatic: 3–176–500 through 3–176–580 for Catheters, Urethral, Rubber, Nelaton; and 3–177–600 through 3–178–570 for Catheters, Urethral, Rubber, Self-Retaining.

6.3 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.4 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of the specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications, of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 Army, Navy, and Air Force.

7.2.1 *Applicable specifications.*

Federal specifications:

NN–B–601—Boxes; Wood - Cleated - Plywood, for Domestic Shipment.

NN–B–621—Boxes; Wood, Nailed and Lock-Corner.

NN–B–631—Boxes; Wood, Wirebound, for Domestic Shipment.

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Military specifications:

JAN–P–105—Packaging and Packing for Overseas Shipment—Boxes, Wood, Cleated, Plywood.

JAN–P–106—Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.

JAN–B–107—Boxes, Wood, Wirebound (Overseas Type).

JAN–P–120—Packaging and Packing for Overseas Shipment—Cartons, Folding, Paperboard.

JAN–P–125—Packaging and Packing for Overseas Shipment—Barrier-Materials, Waterproof, Flexible.

JAN–P–133—Packaging and Packing for Overseas Shipment—Boxes, Set-Up, Paperboard.

JAN–P–140—Packaging and Packing for Overseas Shipment—Adhesives, Water-Resistant, Case-Liner.

MIL–M–3157—Marking of Domestic Shipments by Contractors for Supplies and Equipment Consigned to the Medical Departments of the Armed Services.

Navy Department specification:

General Specifications for Inspection of Material.

(Copies of specifications required for Army purchases may be obtained upon application to the procuring agency. Both the title and identifying symbol or number should be stipulated when requesting copies.

(Copies of Federal and Military specifications required for Navy purchases, and Navy Department specifications, may be obtained without cost upon application to the Bureau of Supplies and Accounts, Department of the Navy, Washington 25, D. C., except that Naval activities should make application to the Commanding Officer, Naval Supply Center, Norfolk 11, Va. Both the title and identifying symbol or number should be stipulated when requesting copies of specifications.

(Copies of specifications required for Air Force purchases may be obtained upon application to the Commanding General, Air Matériel Command, Wright-Patterson Air Force Base, Dayton 1, Ohio. Both the title and identifying symbol or number should be stipulated when requesting copies.)

7.2.2 *Packaging.*

7.2.2.1 *Unit.*—One catheter of the type, style, and composition specified in the purchase order, contract, or invitation for bids constitutes one unit. Each catheter shall be packaged (not coiled) in a waxed paper or glassine envelope and secured. As an alternate to the above, each unit may be completely wrapped (not coiled) in waxed or glassine paper and secured.

7.2.2.2 *Intermediate.*—Six unit packages shall be further packaged in a snug-fitting, folding paperboard carton, or set-up paperboard box conforming to the requirements of Military Specifications JAN–P–120 and JAN–P–133, respectively. The carton shall be adequately secured to prevent accidental opening.

7.2.3 *Packing.*

7.2.3.1 *Domestic.*—Sixty intermediate cartons (360 units) shall be packed in a shipping container conforming to the requirements of Federal Specifications NN–B–601, NN–B–621, or NN–B–631.

7.2.3.2 *Overseas.*—Sixty intermediate cartons (360 units) shall be packed for type 2 load in a shipping container conforming to the requirements of Military Specifications JAN–P–105, JAN–P–106, or JAN–B–107. Closure and strapping shall comply with the appendix to the applicable specification. Each box shall be lined with a sealed waterproof bag or its equivalent, made from material conforming to the requirements of Military Specification JAN–P–125 for case liners; seams and closures shall be completely sealed with a continuous seam at least ¾ inch wide of water-resistant adhesive conforming to the requirements of Military Specification JAN–P–140.

7.2.3.3 *Packing variation permitted.*—If the number of units in any shipment is not the exact multiple of the number of units specified to be contained in each shipping container, the units not constituting the nearest such multiple shall be packed in a shipping container of suitable size and design and so constructed as to insure safe delivery to specified destination.

7.2.4 *Marking.*

7.2.4.1 *Unit packages and intermediate cartons.*—Each unit package and each intermediate carton shall be marked with the stock number, nomenclature, unit, quantity, date of manufacture (month and year), and the name of the contractor. When the contractor is not the

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manufacturer the name of manufacturer shall also appear.

7.2.4.2 *Shipping containers.*—Shipping containers shall be marked in accordance with Military Specification MIL–M–3157.

7.2.5 *Ordering data.*—Requests, requisitions, schedules, and contracts or orders should specify domestic or overseas packing, and marking of shipping containers. (See 7.2.3 and 7.2.4.)

Navy Interest: M.

This specification is a part of Section IV, Part 5, of the Federal Standard Stock Catalog

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